

Matchings and WLP

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In 2023 during a talk, Uwe Nagel posed the following question:

What is the probability that a monomial ideal fails the WLP?

In a later paper, Nagel and Petrović [4] made conjectures that for suitable probabilistic models, the probability of failure should be very small, unless parameters are chosen very carefully. Then, in 2026 we showed [2] that for probabilistic models arising from combinatorics, the algebras defined by monomial ideals of the form $I + (x_1^{d+2}, \dots, x_n^{d+2})$, where I is a squarefree monomial ideal such that $\dim R/I = d + 1$, fail the WLP asymptotically almost surely (for a special choice of parameters).

A special case of Uwe's question is when we restrict ourselves to algebras of the form

$$A(G, d) = \frac{k[x_1, \dots, x_n]}{I(G) + (x_1^d, \dots, x_n^d)}$$

where $I(G)$ is the edge ideal of G , and d is a fixed integer. The closest results to probabilistic-type statements for these algebras can be found in [3], where large classes of graphs are shown to fail the WLP. One missing theoretical aspect to be able to prove probabilistic statements for these algebras is a criterion for failure of WLP that is sufficiently general. With this gap and the tools used in [3] in mind, we propose the following problem:

Problem 1. *Given a graph G , determine criteria for failure of the WLP of $A(G, d)$ by knowing the matching polynomial and independence polynomials of G and its complement.*

A tool we expect to use in the problem above is a family of polynomials called *Specht polynomials*. These polynomials were originally introduced in the context of representation theory, but were later shown to completely characterize the kernel of multiplication maps by (powers of) $x_1 + \dots + x_n$ in the algebra $R/(x_1^2, \dots, x_n^2)$ (see [1, Section 9.3]).

References

- [1] Tadahito Harima, Toshiaki Maeno, Hideaki Morita, Yasuhide Numata, Akihito Wachi, and Junzo Watanabe. *The Lefschetz properties*, volume 2080 of *Lecture Notes in Mathematics*. Springer, Heidelberg, 2013.

- [2] Thiago Holleben. Coinvariant stresses, Lefschetz properties and random complexes. *Adv. in Appl. Math.*, 178:Paper No. 103097, 34, 2026.
- [3] Thiago Holleben and Lisa Nicklasson. Roller coaster Gorenstein algebras and Koszul algebras failing the weak Lefschetz property. *J. Pure Appl. Algebra*, 230(4):Paper No. 108238, 20, 2026.
- [4] Uwe Nagel and Sonja Petrović. The weak lefschetz property and unimodality of hilbert functions of random monomial algebras, 2024.